**Program 5:Demonstrate CI Integration by Running Jenkins in a Docker Container**
The program focuses on demonstrating Continuous Integration(CI) by deploying and running Jenkins inside a Docker container. This setup showcases how Docker simplifies Jenkins installation and enables automated build and integration workflows.

Project Structure:
program_5/
—>Dockerfile
—>docker-compose.yml

Procedure:
Create a folder (ex: program_5)

Step 1: Create a Dockerfile without any extensions and add the following content
nano Dockerfile

1rv24mc074_poornimat@lenovo: ~/p5

GNU nano 7.2 Dockerfile

```
FROM jenkins/jenkins:lts

#Switch to root user to instal the docker tools
USER root

#install the docker inside the jenkins container
RUN apt-get update && apt-get install -y docker.io

#switch to jenkins user
USER jenkins
```

Step 2: Create a docker-compose.yml file with following code
nano docker-compose.yml

```
GNU nano 7.2                                    docker-compose.yml
version: '3.8'

services:
  jenkins:
    build:
      context: .
    container_name: jenkins-ci
    ports:
      - "8080:8080"
      - "50000:50000"
    volumes:
      - jenkins_home:/var/jenkins_home
      - /var/run/docker.sock:/var/run/docker.sock

volumes:
  jenkins_home:

                                        [ Read 17 lines ]
```

Step 3: Build an image

docker build -t program_5 .
Run the container
docker run -p 8080:8080 program_5

```
1rv24mc074_poornimat@lenovo:~/p5$ docker stop jenkins-ci
jenkins-ci
1rv24mc074_poornimat@lenovo:~/p5$ docker rm jenkins-ci
jenkins-ci
1rv24mc074_poornimat@lenovo:~/p5$ docker compose up -d
WARN[0000] /home/1rv24mc074_poornimat/p5/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, pl
ease remove it to avoid potential confusion
[+] Running 1/1
 ✔ Container jenkins-ci  Started                                                                                0.3s
1rv24mc074_poornimat@lenovo:~/p5$
```

Step 4: Verify localhost details on browser
http://localhost:8080